ELEC 231 — Mathematical Methods I for ECE · Fall 2026

Week 5: Homogeneous second order ODEs with constant coefficients

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Typed from Prof. Korenberg's Fall 2025 handwritten notes, version 2026-10-02.

This note covers selected content from Chapters 4–6 of Murray R. Spiegel's *Applied Differential Equations*, and Chapters 8 and 12 of Nilsson & Riedel's *Electric Circuits* (12th ed.).

Notation.

j	imaginary unit, $\mathrm{j}^2=-1$	$\mathcal{L}\{f(t)\}$	Laplace transform
$\mathrm{e}^{-t/\tau}$	natural exponential	$\mathcal{L}^{-1}\{F(s)\}$	inverse transform
$\frac{\mathrm{d}y}{\mathrm{d}t}$, $\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$	derivatives	$v * h$	convolution
$u(t)$	unit step	$\boldsymbol{x}$, $\mathbf{A}$	vector, matrix
$\delta(t)$	unit impulse	τ	time constant

Note (The equation numbers in this unit). The Fall 2025 sheets for this week form one continuously hand-numbered run of ten pages split across three postings. The professor's own equation numbers, however, restart with each topic and collide across files: the numbers (14)–(16) used on hand page ⑤ had already been used for different equations in the previous (exact-ODE) set, and the numbers (9)–(13) never appear at all — the numbering jumps from (8) on page ④ to (14) on page ⑤. The typed notes therefore number every equation continuously as (5.1), (5.2), …; in the draft build each one carries the professor's own number as a margin note beside the display, and every cross-reference in his text ("by (7)", "substituting into (17)") has been resolved to the typed number.

Remark. Hand page ① of this run is not lecture material: it is the professor's worked solution to Tutorial 4 (the exact equation $y^3\,\mathrm{d}t + y^2(3+3t-y)\,\mathrm{d}y = 0$), discussed at the start of the class. See the Tutorial 4 solutions.

1 Natural and step responses of RLC circuits

Source: Nilsson & Riedel, *Electric Circuits*, "Natural and step responses of RLC circuits".

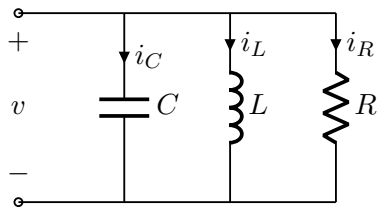


Figure 5.1. Fig. 1 of the notes: the source-free (natural-response) parallel RLC circuit. The branch currents i_C, i_L, i_R all leave the top node, and the port voltage v is positive at the top.

For the circuit of Figure 5.1,

$$i_C = C\frac{\mathrm{d}v}{\mathrm{d}t}, \qquad v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}, \qquad i_R = \frac{v}{R} \tag{5.1}$$

$$i_C + i_L + i_R = 0 \tag{5.2}$$

$$i_L(t) = \frac{1}{L}\int_0^t v(\lambda)\ \mathrm{d}\lambda \tag{5.3}$$

so that Eq. (5.2) becomes

$$\frac{1}{L}\int_0^t v(\lambda)\ \mathrm{d}\lambda + C\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{R} = 0 \tag{5.4}$$

Differentiating Eq. (5.4) with respect to t,

$$\frac{1}{L}\,v(t) + C\frac{\mathrm{d}^2v}{\mathrm{d}t^2} + \frac{1}{R}\frac{\mathrm{d}v}{\mathrm{d}t} = 0,$$

and dividing by C,

$$\frac{\mathrm{d}^2v}{\mathrm{d}t^2} + \frac{1}{RC}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{LC}\,v(t) = 0 \tag{5.5}$$

Equation (5.5) is a linear, second-order homogeneous ODE with constant coefficients.

1.1 Solution

Set $v = A\mathrm{e}^{st}$ in Eq. (5.5):

$$As^2\mathrm{e}^{st} + \frac{1}{RC}\,As\mathrm{e}^{st} + \frac{1}{LC}\,A\mathrm{e}^{st} = 0,$$

which gives

$$s^2 + \frac{1}{RC}\,s + \frac{1}{LC} = 0, \tag{5.6}$$

the *characteristic equation* of a parallel RLC circuit. Equation (5.6) has roots

$$\left.\begin{aligned} s_1 &= -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \\ s_2 &= -\frac{1}{2RC} - \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \end{aligned}\right\} \tag{5.7}$$

The natural response of the parallel RLC circuit in Figure 5.1 is

$$v = A_1\mathrm{e}^{s_1 t} + A_2\mathrm{e}^{s_2 t} \tag{5.8}$$

Regarding Eq. (5.7), three cases of interest arise.

(1) $\dfrac{1}{2RC} > \sqrt{1/(LC)}$.
Here Eq. (5.6) has 2 unequal real roots, and the voltage response v is called *over-damped*. See example in Nilsson & Riedel.

(2) $\dfrac{1}{2RC} = \sqrt{\dfrac{1}{LC}}$.

Here Eq. (5.6) has 2 equal real roots, and voltage response v is called "critically damped"; and voltage response is

$$v(t) = D_1\, t\, \mathrm{e}^{-\alpha t} + D_2\, \mathrm{e}^{-\alpha t}.$$

See example in Spiegel, *Applied Differential Equations*, chapter Applications — Linear Equations: Constant Coefficients.

(3) $\dfrac{1}{2RC} < \sqrt{1/(LC)}$.

Here Eq. (5.6) has 2 complex conjugate roots, and voltage response v is "underdamped". See example in Nilsson & Riedel.

Added explanation (not in the 2025 notes) (the α of case (2)). The symbol α in case (2) is the repeated root of Eq. (5.6): when $1/(2RC) = \sqrt{1/(LC)}$ the square root in Eq. (5.7) vanishes and $s_1 = s_2 = -1/(2RC)$, so $\alpha = 1/(2RC)$. Nilsson & Riedel call α the *neper frequency* and $\omega_0 = \sqrt{1/(LC)}$ the *resonant radian frequency*; the three cases are then simply $\alpha > \omega_0$, $\alpha = \omega_0$ and $\alpha < \omega_0$.

1.2 Existence and uniqueness of solutions

Theorem 5.1 (Existence and uniqueness of solutions). Let $a_0(t)$, $a_1(t)$, …, $a_n(t)$ and $F(t)$ be functions which are continuous in the interval $a \le t \le b$ and suppose that $a_0(t) \neq 0$ for any value of t in the interval. Then there exists a solution $y(t)$ satisfying linear ODE:

$$a_0(t)\frac{\mathrm{d}^n y}{\mathrm{d}t^n} + a_1(t)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}t^{n-1}} + \cdots + a_n(t)\, y = F(t)$$

and also the conditions

$$y(c) = p_0, \quad y'(c) = p_1, \quad y''(c) = p_2, \quad \ldots, \quad y^{(n-1)}(c) = p_{n-1},$$

where $a \le c \le b$, and p_0, p_1, …, p_{n-1} are given constants. Also, this solution is unique.

Source: M. R. Spiegel, *Applied Differential Equations*.

2 Derivation of the roots of the characteristic equation

Derivation of Eq. (5.7) for the roots of characteristic Eq. (5.6). Recall that

$$ax^2 + bx + c_1 = 0 \tag{5.9}$$

has solution

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac_1}}{2a} \tag{5.10}$$

Comparing Eq. (5.9) with Eq. (5.5) yields $a = 1$, $b = 1/RC$, $c_1 = 1/LC$. Hence from Eq. (5.10) we obtain the roots are

$$\begin{aligned} s_{1,2} &= \frac{-\dfrac{1}{RC} \pm \sqrt{\left(\dfrac{1}{RC}\right)^2 - \dfrac{4}{LC}}}{2} \\ &= -\frac{1}{2RC} \pm \sqrt{\frac{1}{4}\left(\frac{1}{RC}\right)^2 - \frac{1}{LC}} \\ &= -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \end{aligned}$$

which is Eq. (5.7).

3 Step response of a parallel RLC circuit

See p. 145 of Spiegel, *Applied Differential Equations*, Fundamental Theorem 2.

Theorem 5.2 (Fundamental Theorem 2). The general solution of

$$a_0(t)\frac{\mathrm{d}^n y}{\mathrm{d}t^n} + a_1(t)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}t^{n-1}} + \cdots + a_{n-1}(t)\frac{\mathrm{d}y}{\mathrm{d}t} + a_n(t)\, y = F(t) \tag{5.11}$$

may be obtained by finding a particular solution y_P of this equation, and adding it to the general solution of Eq. (5.11) with $F(t) \equiv 0$.

Source: M. R. Spiegel, *Applied Differential Equations*, p. 145.

Example 5.1 (Nilsson & Riedel). Find the step response of a parallel RLC circuit.

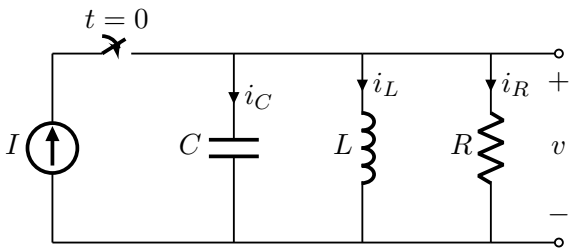


Figure 5.2. Fig. 1 of the step-response example (the professor reuses the label "Fig. 1"): a dc current source I is switched onto the parallel combination of C, L and R at $t = 0$.

Summing the currents i_C, i_L, i_R, and I into the top node yields

$$i_C + i_L + i_R = I, \tag{5.12}$$

where

$$i_C = C\frac{\mathrm{d}v}{\mathrm{d}t}, \qquad v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}, \qquad i_R = \frac{v}{R}.$$

Hence

$$i_C = CL\frac{\mathrm{d}^2 i_L}{\mathrm{d}t^2}, \qquad i_R = \frac{L}{R}\frac{\mathrm{d}i_L}{\mathrm{d}t},$$

so substituting into Eq. (5.12) yields

$$CL\frac{\mathrm{d}^2 i_L}{\mathrm{d}t^2} + \frac{L}{R}\frac{\mathrm{d}i_L}{\mathrm{d}t} + i_L = I.$$

Dividing by LC, we obtain

$$\frac{\mathrm{d}^2 i_L}{\mathrm{d}t^2} + \frac{1}{RC}\frac{\mathrm{d}i_L}{\mathrm{d}t} + \frac{1}{LC}\,i_L = \frac{1}{LC}\,I \tag{5.13}$$

To invoke Fundamental Theorem 2 (Theorem 5.2), we need a particular solution I_P of Eq. (5.13). Clearly, if $I_P = I$, a constant, it satisfies Eq. (5.13), because then

$$\frac{\mathrm{d}I_P}{\mathrm{d}t} = \frac{\mathrm{d}^2 I_P}{\mathrm{d}t^2} = 0.$$

The general solution of Eq. (5.13) when the right side is zero is the natural response of the parallel LRC circuit. Hence the general solution of Eq. (5.13) is I + Natural Response, as noted by Nilsson & Riedel.

3.1 A worked example

Example 5.2 (Nilsson & Riedel). The initial energy stored in the circuit of Figure 5.3 is zero. At $t = 0$, a dc current source of 24 mA is applied to the circuit. $R = 400\,\Omega$. Find $i_L(t)$, $t \geq 0$.

3.1.1 Solution

The energy stored in the capacitor is $W_C = \frac{1}{2}Cv^2$. The energy stored in the inductor is $W_L = \frac{1}{2}L\,i_L^2$. Hence, zero initial energy stored in the circuit implies

$$v(0) = 0, \qquad i_L(0) = 0.$$

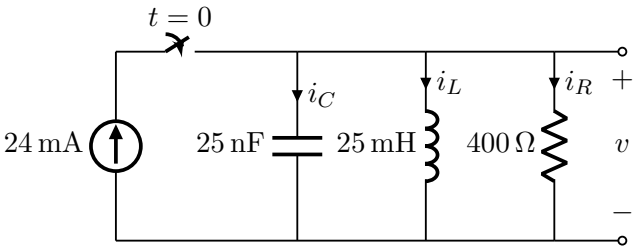


Figure 5.3. Fig. 2 of the notes: the circuit of the worked example, with $C = 25\,\mathrm{nF}$, $L = 25\,\mathrm{mH}$, $R = 400\,\Omega$ and a 24 mA dc current source switched on at $t = 0$.

$$\frac{1}{2RC} = \frac{1}{2 \times 400 \times 25 \times 10^{-9}} = 50\,000\,\mathrm{rad/s}$$

$$\sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{25 \times 10^{-3} \times 25 \times 10^{-9}}} = 40\,000\,\mathrm{rad/s}$$

Since $\sqrt{1/LC} < 1/(2RC)$, the roots s_1, s_2 are real and unequal, by Eq. (5.7)[C1], so the step response is over-damped.

To get the natural response of the parallel LRC circuit we solve the characteristic equation for roots s_1, s_2 when the right side of Eq. (5.13) is zero:

$$s^2 + \frac{1}{RC}\,s + \frac{1}{LC} = 0.$$

By Eq. (5.7),

$$\begin{aligned} s_{1,2} &= -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \\ &= -50\,000 \pm \sqrt{(50\,000)^2 - (40\,000)^2} \\ &= -50\,000 \pm 30\,000 \end{aligned}$$

so that

$$s_1 = -20\,000\,\text{rad/s}, \qquad s_2 = -80\,000\,\text{rad/s}.$$

Hence

$$i_L(t) = \left(24 + A_1 \mathrm{e}^{-20\,000\,t} + A_2 \mathrm{e}^{-80\,000\,t}\right)\ \text{mA}.$$

To find A_1, A_2, we need $i_L(0)$ and $\frac{\mathrm{d}i_L}{\mathrm{d}t}\Big|_{t=0}$. Because no initial energy is stored in circuit, $i_L(0) = 0$ and $v(0) = 0$, so that $L\frac{\mathrm{d}i_L}{\mathrm{d}t}\Big|_{t=0} = 0$. Therefore

$$\begin{aligned} i_L(0) = 0 &\implies 24 + A_1 + A_2 = 0 \implies A_1 + A_2 = -24\,\text{mA}, \\ \left.\frac{\mathrm{d}i_L(t)}{\mathrm{d}t}\right|_{t=0} = 0 &\implies -20\,000\,A_1 - 80\,000\,A_2 = 0. \end{aligned}$$

Solving yields

$$A_1 = -32\,\text{mA}, \qquad A_2 = 8\,\text{mA}.$$

Added explanation (not in the 2025 notes) (the two-by-two system). The professor writes only "solving yields". From the second equation $-20\,000A_1 = 80\,000A_2$, i.e. $A_1 = -4A_2$; substituting into $A_1 + A_2 = -24\,\text{mA}$ gives $-3A_2 = -24$, so $A_2 = 8\,\text{mA}$ and $A_1 = -32\,\text{mA}$.

Hence

Step response of the parallel RLC circuit.

$$i_L(t) = \left(24 - 32\,\mathrm{e}^{-20\,000\,t} + 8\,\mathrm{e}^{-80\,000\,t}\right)\ \text{mA}, \qquad t \geq 0.$$

[C1] Corrected from the handwritten original, which reads: "by eq (9)" — equations (9)–(13) never appear in the 2025 scans, and the root formula being quoted is the professor's Eq. (7). Source: Parallel_LRC:4.

4 Heaviside step function

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases}$$

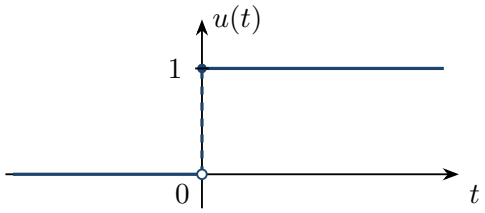


Figure 5.4. The Heaviside step function $u(t)$, as sketched in the notes: zero before the origin, unity from $t = 0$ onwards.

5 Reduction of order

Theorem 5.3 (Reduction of order). If $y = y_1$ is a known solution of the linear n-th order equation

$$a_0(t)\frac{\mathrm{d}^n y}{\mathrm{d}t^n} + a_1(t)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}t^{n-1}} + a_2(t)\frac{\mathrm{d}^{n-2} y}{\mathrm{d}t^{n-2}}^{\mathrm{C2}} + \cdots + a_{n-1}(t)\frac{\mathrm{d}y}{\mathrm{d}t} + a_n(t)\, y^{\mathrm{C3}} = F(t)$$

then substituting $y = y_1 v$ will transform the equation into an equation of $(n-1)$st order in $v' = \frac{\mathrm{d}v}{\mathrm{d}t}$.

Source: M. R. Spiegel, *Applied Differential Equations*, p. 151.

Added explanation (not in the 2025 notes) (where this is used). Reduction of order is carried out on a concrete example at the start of Week 6: $y'' - t\,y' + y = 0$, given the known solution $y = t$.

[C2] Corrected from the handwritten original, which reads: "$a_2(t)\ \mathrm{d}^{n-2}y/\,\mathrm{d}t^{n-1}$" — in the Spiegel p. 151 theorem the k-th term is $a_k(t)\ \mathrm{d}^{n-k}y/\,\mathrm{d}t^{n-k}$, so the derivative order in numerator and denominator must agree. Source: Parallel_LRC_problem:2.

[C3] Corrected from the handwritten original, which reads: "$a_0(t)\,y$" — $a_0(t)$ already multiplies the highest derivative in the same displayed equation, so the coefficient of the undifferentiated y must be $a_n(t)$. Source: Parallel_LRC_problem:2.